

PH102: Tutorial 12 solution

1. Let us first calculate the expression for the intensity of light using Maxwell's equation. Assume

$$\vec{E} = \hat{x} E_0 e^{i(kz - \omega t)}$$

$$\vec{B} = \frac{\vec{k} \times \vec{E}}{\omega} = \hat{y} \frac{k E_0}{\omega} e^{i(kz - \omega t)}$$

$$\vec{S} = \vec{E} \times \vec{H}$$

$$= \hat{z} \frac{k E_0^2}{\mu_0 \omega} e^{2i(kz - \omega t)}$$

. Note that we generally consider $\mu = \mu_0$.

$$\langle \vec{S} \rangle = I = \frac{1}{2} \frac{k E_0^2}{\omega \mu_0}$$

$$k = \frac{\omega}{v} = \omega \sqrt{\epsilon \mu_0}$$

Thus we have

$$I = \frac{1}{2} \sqrt{\frac{\epsilon}{\mu_0}} E_0^2$$

$$\Rightarrow I \propto \sqrt{\epsilon} E_0^2$$

As I is same in the air and dielectric we obtain

$$\sqrt{\epsilon_{air}} E_{0(air)}^2 = \sqrt{\epsilon_r} E_{0(dielectric)}^2$$

$$\Rightarrow E_{0(dielectric)} = \left(\sqrt{\frac{\epsilon_{air}}{\epsilon_r}} \right)^{1/2} E_{0(air)}$$

$$= \left(\frac{1}{10.9} \right)^{1/4} 4 \times 10^9 = 2.2 \times 10^9 \text{ V/m}$$

2. (a)

$$\nabla \times \mathbf{E} = \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta E_\phi) \hat{r} - \frac{1}{r} \frac{\partial}{\partial r} (r E_\phi) \hat{\theta}$$

Let's consider $u = kr - \omega t$ Therefore we will have

$$= \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} \left[E_0 \frac{\sin^2 \theta}{r} \left(\cos u - \frac{1}{kr} \sin u \right) \right] \hat{r} - \frac{1}{r} \frac{\partial}{\partial r} \left[E_0 \sin \theta \left(\cos u - \frac{1}{kr} \sin u \right) \right] \hat{\theta}$$

we have

$$\frac{\partial}{\partial r} \cos u = -k \sin u; \frac{\partial}{\partial r} \sin u = k \cos u.$$

We obtain,

$$= \frac{1}{r \sin \theta} \frac{E_0}{r} 2 \sin \theta \cos \theta \left(\cos u - \frac{1}{kr} \sin u \right) \hat{r} - \frac{1}{r} E_0 \sin \theta \left(-k \sin u + \frac{1}{kr^2} \sin u - \frac{1}{r} \cos u \right) \hat{\theta}$$

Integrating with respect to t , and noting that $\int \cos u dt = -\frac{1}{\omega} \sin u$ and $\int \sin u dt = \frac{1}{\omega} \cos u$

We finally get

$$\boxed{\mathbf{B} = \frac{2E_0 \cos \theta}{\omega r^2} \left(\sin u + \frac{1}{kr} \cos u \right) \hat{r} + \frac{E_0 \sin \theta}{\omega r} \left(-k \cos u + \frac{1}{kr^2} \cos u + \frac{1}{r} \sin u \right) \hat{\theta}}$$

(b) Poynting vector

$$\begin{aligned} \mathbf{S} &= \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{E_0 \sin \theta}{\mu_0 r} \left(\cos u - \frac{1}{kr} \sin u \right) \left[\frac{2E_0 \cos \theta}{\omega r^2} \left(\sin u + \frac{1}{kr} \cos u \right) \hat{\theta} \right] \\ &+ \frac{E_0 \sin \theta}{\mu_0 r} \left(\cos u - \frac{1}{kr} \sin u \right) \left[\frac{E_0 \sin \theta}{\omega r} \left(-k \cos u + \frac{1}{kr^2} \cos u + \frac{1}{r} \sin u \right) (-\hat{r}) \right] \\ &= \frac{E_0^2 \sin \theta}{\mu_0 \omega r^2} \left[\frac{2 \cos \theta}{r} \left(1 - \frac{1}{k^2 r^2} \right) \sin u \cos u + \frac{1}{kr} (\cos^2 u - \sin^2 u) \right] \hat{\theta} \\ &+ \sin \theta \left[\left(-\frac{2}{r} + \frac{1}{k^2 r^3} \right) \sin u \cos u + k \cos^2 u + \frac{1}{kr^2} (\sin^2 u - \cos^2 u) \right] \hat{r} \end{aligned}$$

Averaging over the full cycle, using $\langle \sin u \cos u \rangle = 0$, $\langle \cos^2 u \rangle = \frac{1}{2}$ we get the intensity as

$$I = \langle S \rangle = \frac{E_0^2 \sin \theta}{\mu_0 \omega r^2} \left(\frac{k}{2} \sin \theta \right) \hat{r} = \boxed{\frac{E_0^2 \sin^2 \theta}{2\mu_0 c r^2} \hat{r}}$$

It will point towards the $\hat{r}$ direction, and falls off as $1/r^2$, it would be expected for the spherical wave.

$$P = \int \mathbf{I} \cdot d\mathbf{a} = \frac{E_0^2}{2\mu_0 c} \int \frac{\sin^2 \theta}{r^2} r^2 \sin \theta d\theta d\phi = \frac{E_0^2}{2\mu_0 c} 2\pi \int_0^\pi \sin^3 \theta d\theta = \boxed{\frac{4\pi}{3} \frac{E_0^2}{\mu_0 c}}$$

3. (a)

$$\vec{E}_1 = \hat{y} E_{10}^+ e^{i(k_1 x - \omega t)} + \hat{y} E_{10}^- e^{-i(k_1 x + \omega t)}$$

$$\vec{E}_2 = \hat{y}E_{20}^+e^{i(k_2x-\omega t)} + \hat{y}E_{20}^-e^{-i(k_2x+\omega t)}$$

$$\vec{E}_3 = \hat{y}E_{30}^+e^{i(k_3x-\omega t)}$$

(b) Using

$$\vec{H} = \frac{\vec{k} \times \vec{E}}{\omega\mu}$$

Consider $\mu = \mu_0$ for non-magnetic media.

$$\vec{H}_1 = \hat{z}\frac{k_1}{\omega\mu_0} [E_{10}^+e^{i(k_1x-\omega t)} - E_{10}^-e^{-i(k_1x+\omega t)}]$$

$$\vec{H}_2 = \hat{z}\frac{k_2}{\omega\mu_0} [E_{20}^+e^{i(k_2x-\omega t)} - E_{20}^-e^{-i(k_2x+\omega t)}]$$

$$\vec{H}_3 = \hat{z}\frac{k_3}{\omega\mu_0} [E_{30}^+e^{i(k_3x-\omega t)}] \quad \text{No backward propagating wave!!}$$

(c) To make the analysis less rigorous let us introduce a phase factor e^{-ik_3d} in $\vec{E}_3$ and $\vec{H}_3$. Please note that one can proceed without adopting this simplification also.

$$\vec{E}_3 = \hat{y}E_{30}^+e^{i[k_3(x-d)-\omega t]}$$

$$\vec{H}_3 = \hat{z}\frac{k_3}{\omega\mu_0} [E_{30}^+e^{i[k_3(x-d)-\omega t]}]$$

Now tangential component of both E_y and H_z should be continuous at the interface $x = 0$ and $x = d$.

Continuity condition at $x = 0$ gives

$$E_{10}^+ + E_{10}^- = E_{20}^+ + E_{20}^- \quad [1]$$

And

$$\frac{k_1}{\omega\mu_0} [E_{10}^+ - E_{10}^-] = \frac{k_2}{\omega\mu_0} [E_{20}^+ - E_{20}^-] \quad [2]$$

$$k_1 = \frac{\omega}{c}n_1; k_2 = \frac{\omega}{c}n_2$$

So,

$$E_{10}^+ - E_{10}^- = \frac{n_1}{n_2} (E_{20}^+ - E_{20}^-) \quad [3]$$

$$[1] + [3] \implies$$

$$2E_{10}^+ = \left(\frac{n_2}{n_1} + 1\right) E_{20}^+ + \left(1 - \frac{n_2}{n_1}\right) E_{20}^- \quad [4]$$

$$\implies \boxed{E_{10}^+ = \frac{n_1 + n_2}{2n_1} E_{20}^+ + \frac{n_1 - n_2}{2n_1} E_{20}^-}$$

$$[1] - [3] \implies$$

$$2E_{10}^- = \left(1 - \frac{n_2}{n_1}\right) E_{20}^+ + \left(1 + \frac{n_2}{n_1}\right) E_{20}^- \quad [5]$$

$$\implies \boxed{E_{10}^- = \frac{n_1 - n_2}{2n_1} E_{20}^+ + \frac{n_1 + n_2}{2n_1} E_{20}^-}$$

Again continuity condition at $x = d$ gives

$$E_{20}^+ e^{ik_2 d} + E_{20}^- e^{-ik_2 d} = E_{30}^+ \quad [6]$$

$$\frac{k_2}{\omega \mu_0} [E_{20}^+ e^{ik_2 d} - E_{20}^- e^{-ik_2 d}] = \frac{k_3}{\omega \mu_0} E_{30}^+ \quad [7]$$

or

$$E_{20}^+ e^{ik_2 d} - E_{20}^- e^{-ik_2 d} = \frac{n_3}{n_2} E_{30}^+ \quad [8]$$

[6] + [8] will give

$$2E_{20}^+ e^{ik_2 d} = \left(1 + \frac{n_3}{n_2}\right) E_{30}^+$$

$$\implies \boxed{E_{20}^+ = \frac{n_2 + n_3}{2n_2} E_{30}^+ e^{-ik_2 d}} \quad [9]$$

[6] - [8] will give

$$2E_{20}^- = \left(\frac{n_2 - n_3}{2n_2}\right) E_{30}^+ e^{ik_2 d} \quad [10]$$

From [9] and [10] we get

$$E_{20}^+ = \frac{n_2 + n_3}{2n_2} e^{-ik_2 d} \frac{2n_2}{n_2 - n_3} e^{-ik_2 d} E_{20}^-$$

$$\boxed{E_{20}^+ = \frac{n_2 + n_3}{n_2 - n_3} e^{-2ik_2 d} E_{20}^-} \quad [11]$$

Defining

$$r_1 = \frac{n_1 - n_2}{n_2 + n_1}$$

and

$$r_2 = \frac{n_2 - n_3}{n_2 + n_3} \quad \text{and putting [11] in [4] and [5] we obtain}$$

$$\begin{aligned} E_{10}^+ &= \frac{n_1 + n_2}{2n_1} \frac{n_2 + n_3}{n_2 - n_3} e^{-2ik_2 d} E_{20}^- + \frac{n_1 - n_2}{2n_1} E_{20}^- \\ &= \frac{n_1 - n_2}{2n_1} \left[\frac{n_1 + n_2}{n_1 - n_2} \frac{n_2 + n_3}{n_2 - n_3} e^{-2ik_2 d} + 1 \right] \\ &= \frac{n_1 - n_2}{2n_2} \left[\frac{1}{r_1 r_2} e^{-2ik_2 d} + 1 \right] E_{20}^- \end{aligned}$$

Again,

$$\begin{aligned} E_{10}^- &= \frac{n_1 - n_2}{2n_1} \frac{n_2 + n_3}{n_2 - n_3} e^{-i2k_2 d} E_{20}^- + \frac{n_1 + n_2}{2n_1} E_{20}^- \\ &= \frac{n_1 + n_2}{2n_1} \left[\frac{n_1 - n_2}{n_1 + n_2} \frac{n_2 + n_3}{n_2 - n_3} e^{-2ik_2 d} + 1 \right] E_{20}^- \\ &= \frac{n_1 + n_2}{2n_1} \left[\frac{r_1}{r_2} e^{-2ik_2 d} + 1 \right] E_{20}^- \end{aligned}$$

$$r = \frac{E_{10}^-}{E_{10}^+} = \frac{n_1 + n_2}{n_1 - n_2} \frac{(r_1/r_2)e^{-2i\delta} + 1}{[1/(r_1 r_2)]e^{-2i\delta} + 1} \quad \text{here, } \delta = 2k_2 d$$

$$= \frac{1}{r_1} \left[\frac{r_1^2 e^{-2i\delta} + r_1 r_2}{e^{-2i\delta} + r_1 r_2} \right]$$

$$= \frac{r_1 e^{-2i\delta} + r_2}{r_1 r_2 + e^{-2i\delta}}$$

$$r = \frac{r_1 e^{-i\delta} + r_2 e^{i\delta}}{e^{-i\delta} + r_1 r_2 e^{i\delta}}$$

$$R = |r|^2 = \frac{r_1^2 + r_2^2 + 2r_1 r_2 \cos 2\delta}{1 + r_1^2 r_2^2 + 2r_1 r_2 \cos 2\delta}$$

For $\delta = 0$ we have

$$R = \frac{(r_1 + r_2)^2}{1 + r_1^2 r_2^2 + 2r_1 r_2}$$

Putting

$$r_1 = \frac{n_1 - n_2}{n_1 + n_2}$$

and

$$r_2 = \frac{n_2 - n_3}{n_2 + n_3}$$

One can easily obtain after performing some easy algebra

$$R = \left(\frac{n_1 - n_3}{n_1 + n_3} \right)^2$$

4. Energy density is given by

$$u = \frac{1}{2} \left(\epsilon E^2 + \frac{1}{\mu} B^2 \right) = \frac{1}{2} e^{-2kz} \left[\epsilon E_0^2 \cos^2(kz - \omega t + \delta_E) + \frac{1}{\mu} B_0^2 \cos^2(kz - \omega t + \delta_E + \phi) \right]$$

Averaging over the full cycle (using $\langle \cos^2 \theta \rangle = 1/2$) we get

$$\langle u \rangle = \frac{1}{2} e^{-2kz} \left[\frac{\epsilon}{2} E_0^2 + \frac{1}{2\mu} B_0^2 \right] = \frac{1}{4} e^{-2kz} \left[\epsilon E_0^2 + \frac{1}{\mu} E_0^2 \epsilon \mu \sqrt{1 + \left(\frac{\sigma}{\epsilon \omega} \right)^2} \right] = \frac{1}{4} e^{-2kz} \epsilon E_0^2 \left[1 + \sqrt{1 + \left(\frac{\sigma}{\epsilon \omega} \right)^2} \right]$$

Use the relation (Griffith 3rd edition Eq. 9.126)

$$1 + \sqrt{1 + \left(\frac{\sigma}{\epsilon \omega} \right)^2} = \frac{2}{\epsilon \mu} \frac{k^2}{\omega^2}$$

Please note that we will do the above relation in detail (modified wave vector for the conducting media) in the next class on Wednesday (24th April). However there is no tutorial after the class so we have given this problem to have the feel. so,

$$\langle u \rangle = \frac{1}{4} e^{-2kz} \epsilon E_0^2 \frac{2}{\epsilon \mu} \frac{k^2}{\omega^2} = \boxed{\frac{k^2}{2\mu\omega^2} E_0^2 e^{-2kz}}$$

Therefore the ratio of the magnetic and electric energy is

$$\frac{\langle u_{mag} \rangle}{\langle u_{elec} \rangle} = \frac{B_0^2/\mu}{E_0^2\epsilon} = \frac{1}{\mu\epsilon} \mu\epsilon \sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2} > 1$$

5. The electric field associated with the transmitted wave is

$$\vec{E}_2 = \vec{E}_{20} e^{i[k_{2x}x + k_{2z}z - \omega t]}$$

$$k_{2x} = k_2 \cos \theta_2$$

$$k_{2z} = k_2 \sin \theta_2 \quad \text{see fig.}$$

Considering the media to be non-magnetic we have

$$\frac{\sin \theta_2}{\sin \theta_1} = \sqrt{\frac{\epsilon_1}{\epsilon_2}} \implies \cos \theta_2 = \sqrt{1 - \sin^2 \theta_2} = \sqrt{1 - \frac{\epsilon_1}{\epsilon_2} \sin^2 \theta_1}$$

$$\cos \theta_2 = \sqrt{\frac{\epsilon_1}{\epsilon_2}} \sqrt{\frac{\epsilon_2}{\epsilon_1} - \sin^2 \theta_1}$$

$$\sin \theta_c = \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

$$\cos \theta_2 = \sqrt{\frac{\epsilon_1}{\epsilon_2}} \sqrt{\sin^2 \theta_c - \sin^2 \theta_1}$$

For total internal reflection $\theta_1 > \theta_c$. So we can write

$$\cos \theta_2 = \sqrt{\frac{\epsilon_1}{\epsilon_2}} i\alpha$$

with $\alpha = \sqrt{\sin^2 \theta - \sin^2 \theta_c}$ Thus we can write

$$\vec{E}_2 = \vec{E}_{20} e^{i[k_2 \cos \theta_2 x + k_2 \sin \theta_2 z - \omega t]}$$

$$= \vec{E}_{20} e^{i \left[k_2 \sqrt{\frac{\epsilon_1}{\epsilon_2}} i\alpha x + k_2 \sin \theta_2 z - \omega t \right]}$$

$$= \vec{E}_{20} e^{-k_2 \sqrt{\frac{\epsilon_1}{\epsilon_2}} x} e^{i \left[k_2 \sqrt{\frac{\epsilon_1}{\epsilon_2}} \sin \theta_1 z - \omega t \right]}$$

$$= \vec{E}_{20} e^{-\beta x} e^{i \left[k_2 \sqrt{\frac{\epsilon_1}{\epsilon_2}} \sin \theta_1 z - \omega t \right]}$$

first term in the above represents that the amplitude of the wave is decreasing exponentially in the x-direction and second term indicates that the wave is propagating along the z-direction. Such a wave is known as evanescent wave. These waves have tremendous application particularly in the sensing!